

Current Electricity

Chapter [3]

Class - XII

Manish Joshi

D.A.V. Lohaghat

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Limitations of Ohm's Law

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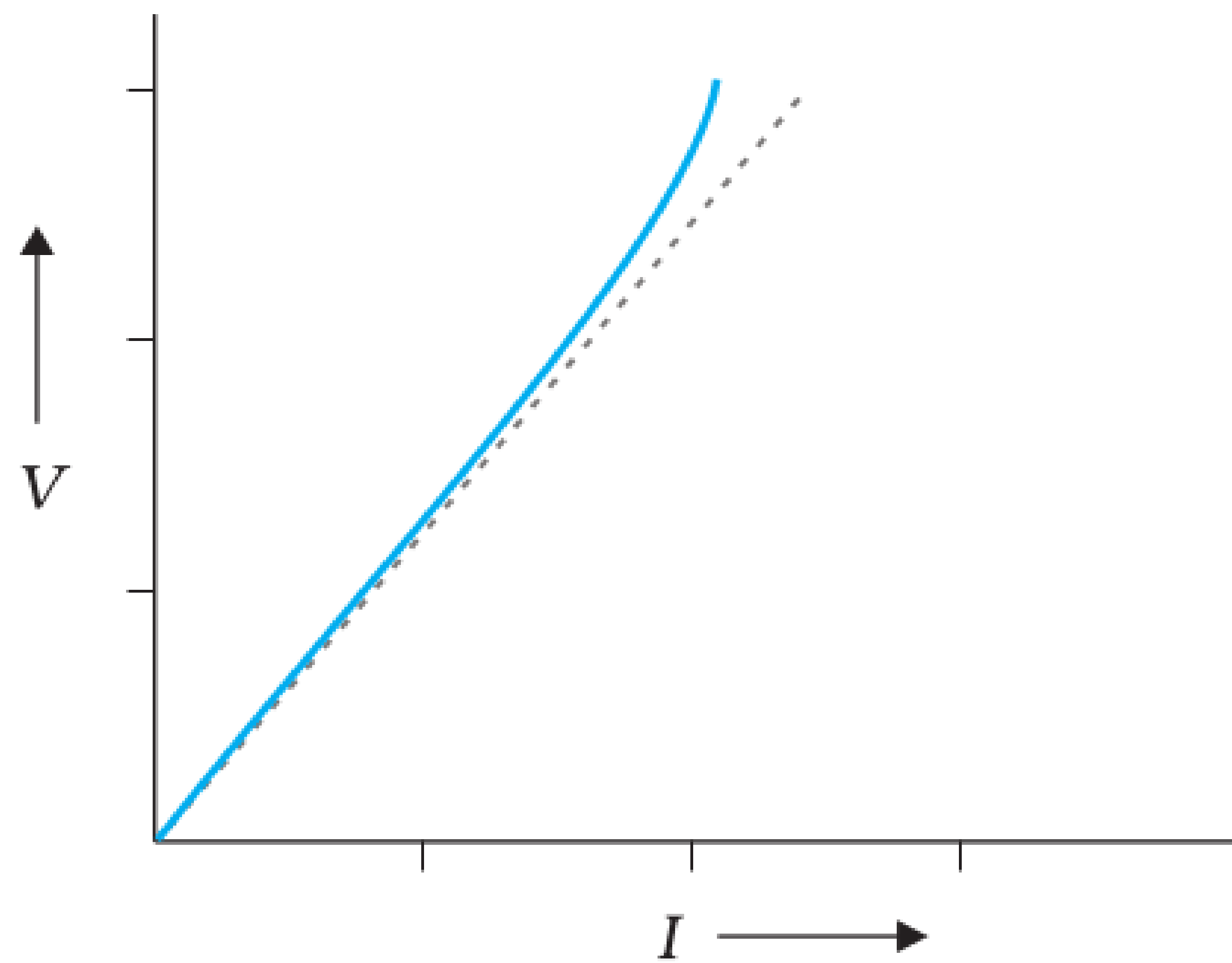


FIGURE 3.5 The dashed line represents the linear Ohm's law. The solid line is the voltage V versus current I for a good conductor.

- But, there do exist materials and devices in which Ohm's law doesn't hold (*i.e.*, $V \propto I$)
 - a V is not proportional to I for some conductor at high voltage.

Limitations of Ohm's Law

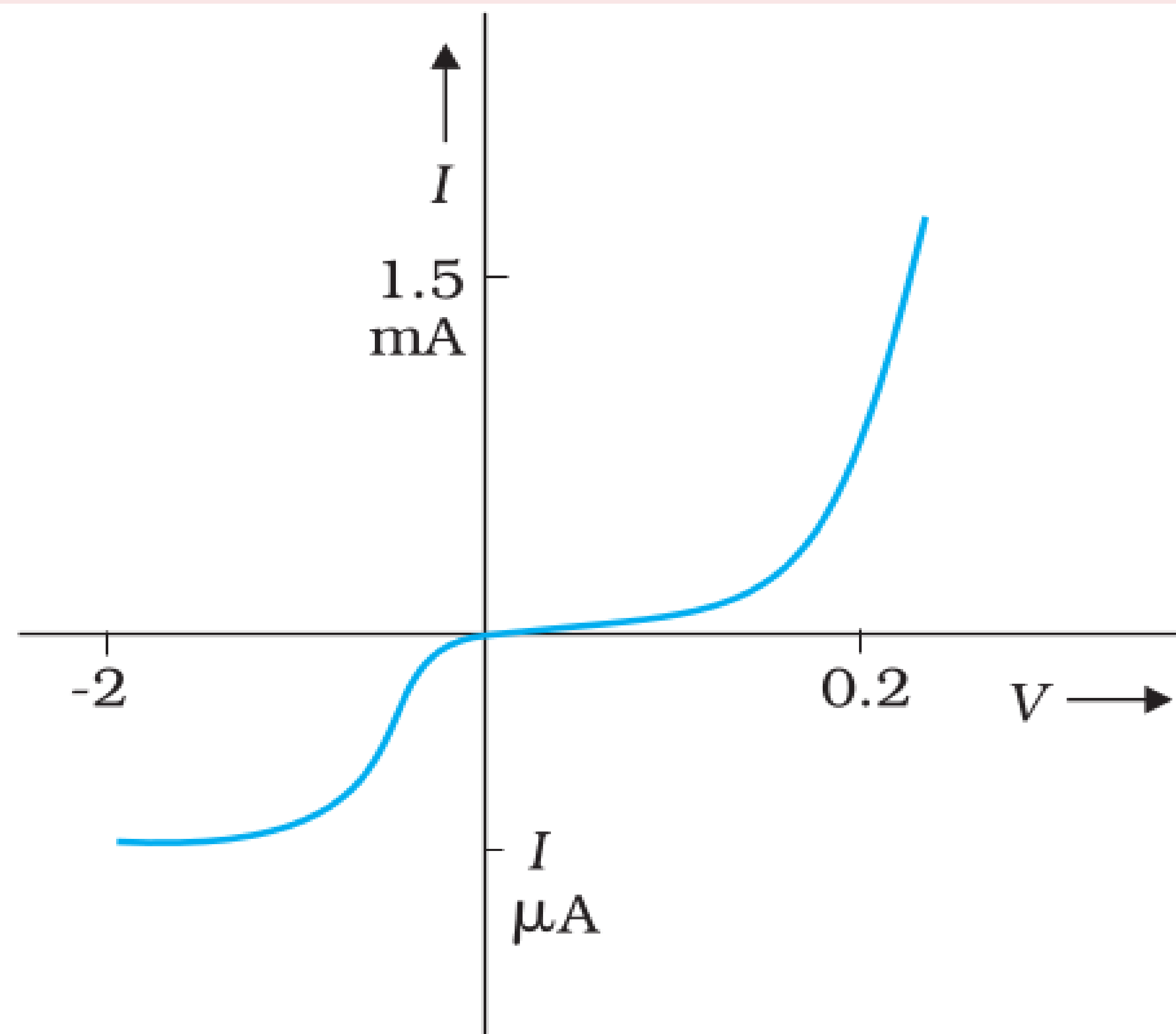


FIGURE 3.6 Characteristic curve of a diode. Note the different scales for negative and positive values of the voltage and current.

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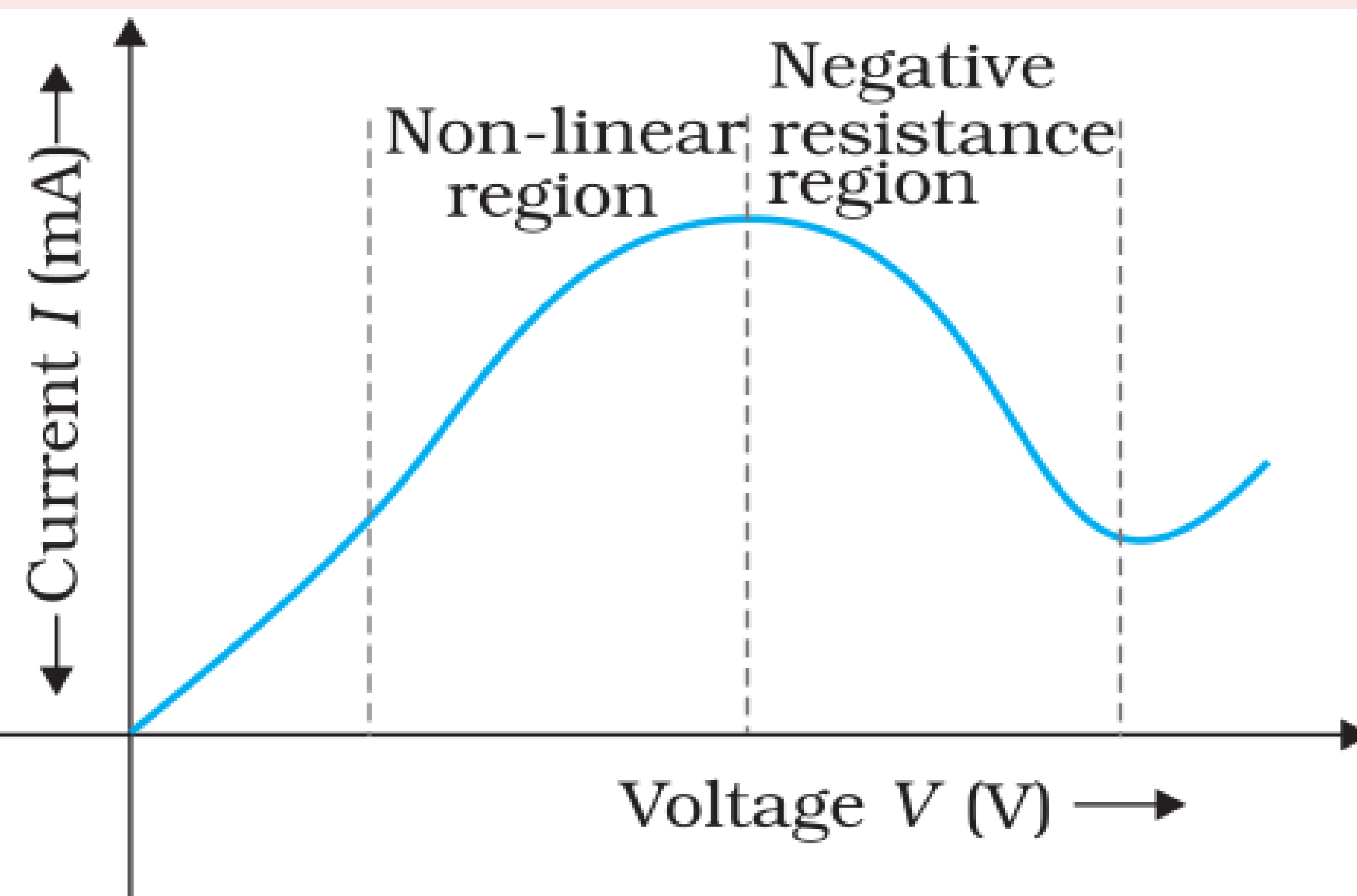


FIGURE 3.7 Variation of current versus voltage for GaAs.

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 - a V is not proportional to I for some conductor at high voltage.
 - b For semiconductors (diodes) the relation between V and I depends on the sign of V .
 - c The relation between V and I is not unique, *i.e.*, there is more than one value of V for the same current I . A material exhibiting such behaviour is GaAs.

RESISTIVITY OF VARIOUS MATERIALS

Material	Resistivity, ρ (Ω m) at 0°C
Conductors	
Silver	1.6×10^{-8}
Copper	1.7×10^{-8}
Aluminium	2.7×10^{-8}
Tungsten	5.6×10^{-8}
Iron	10×10^{-8}
Platinum	11×10^{-8}
Mercury	98×10^{-8}
Nichrome (alloy of Ni, Fe, Cr)	$\sim 100 \times 10^{-8}$
Manganin (alloy)	48×10^{-8}
Semiconductors	
Carbon (graphite)	3.5×10^{-5}
Germanium	0.46
Silicon	2300
Insulators	
Pure Water	2.5×10^5
Glass	$10^{10} - 10^{14}$
Hard Rubber	$10^{13} - 10^{16}$
NaCl	$\sim 10^{14}$
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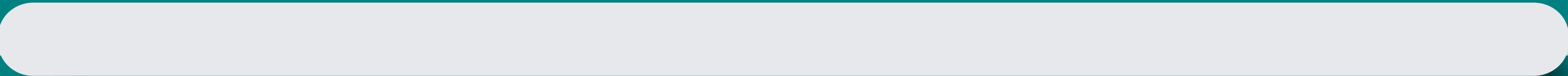
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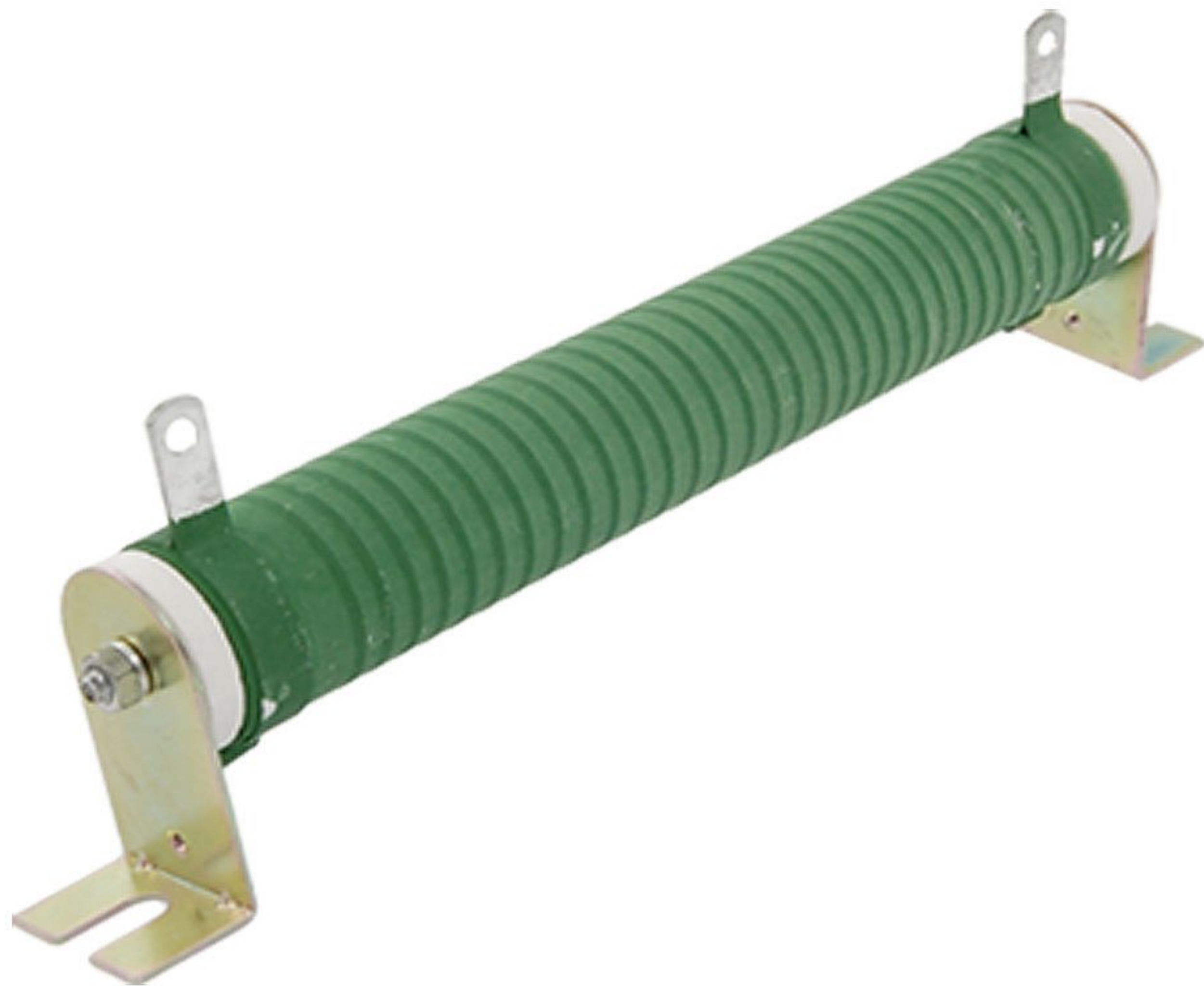
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- **semiconductors** (like Silicon, Germanium etc) have resistivities **in between** the **Conductors** and **Insulators**.

Types of Resistors



There are mainly **two types of Resistors** used in circuits and laboratories :

Types of Resistors

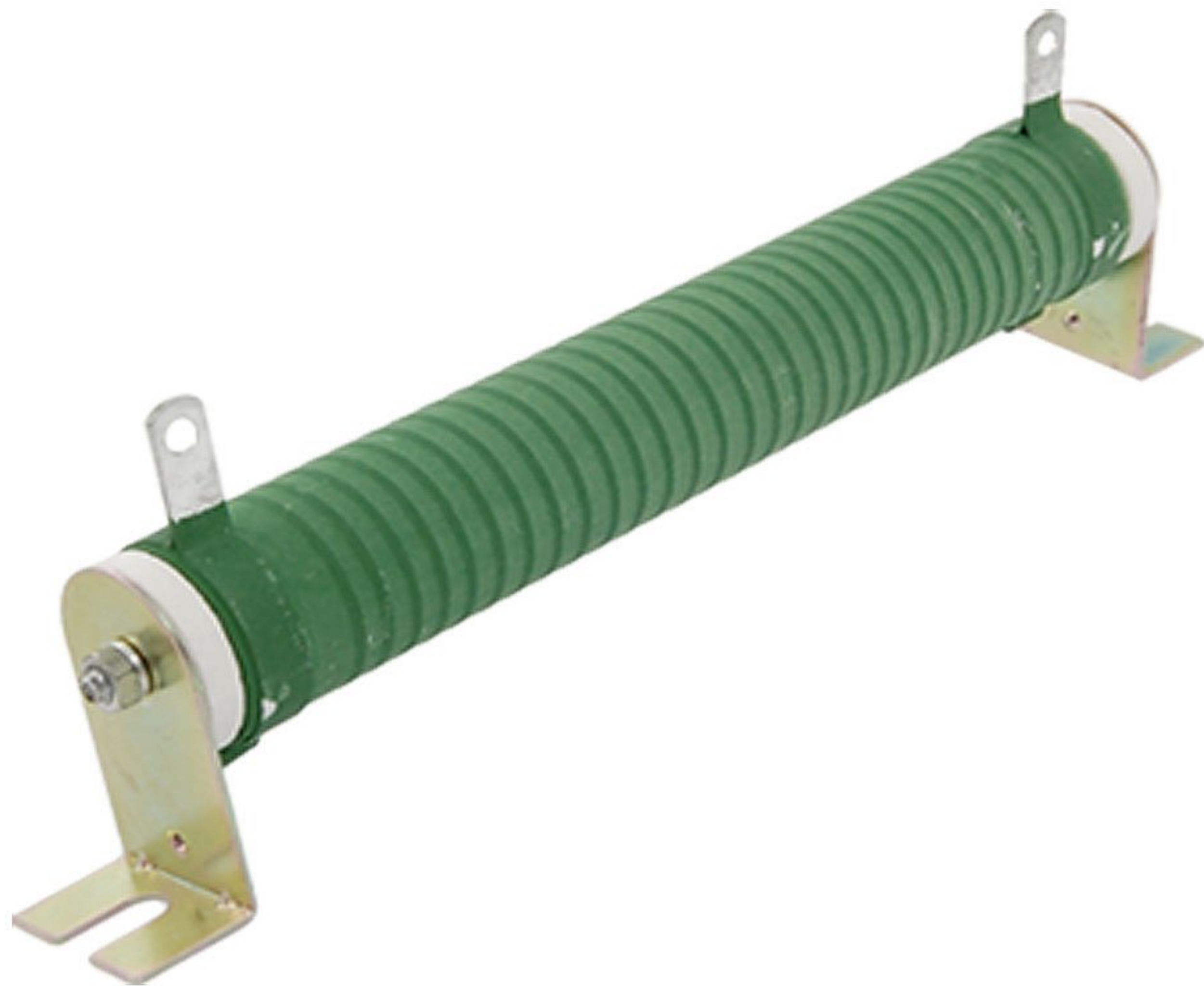


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① **Wire bound resistors :**

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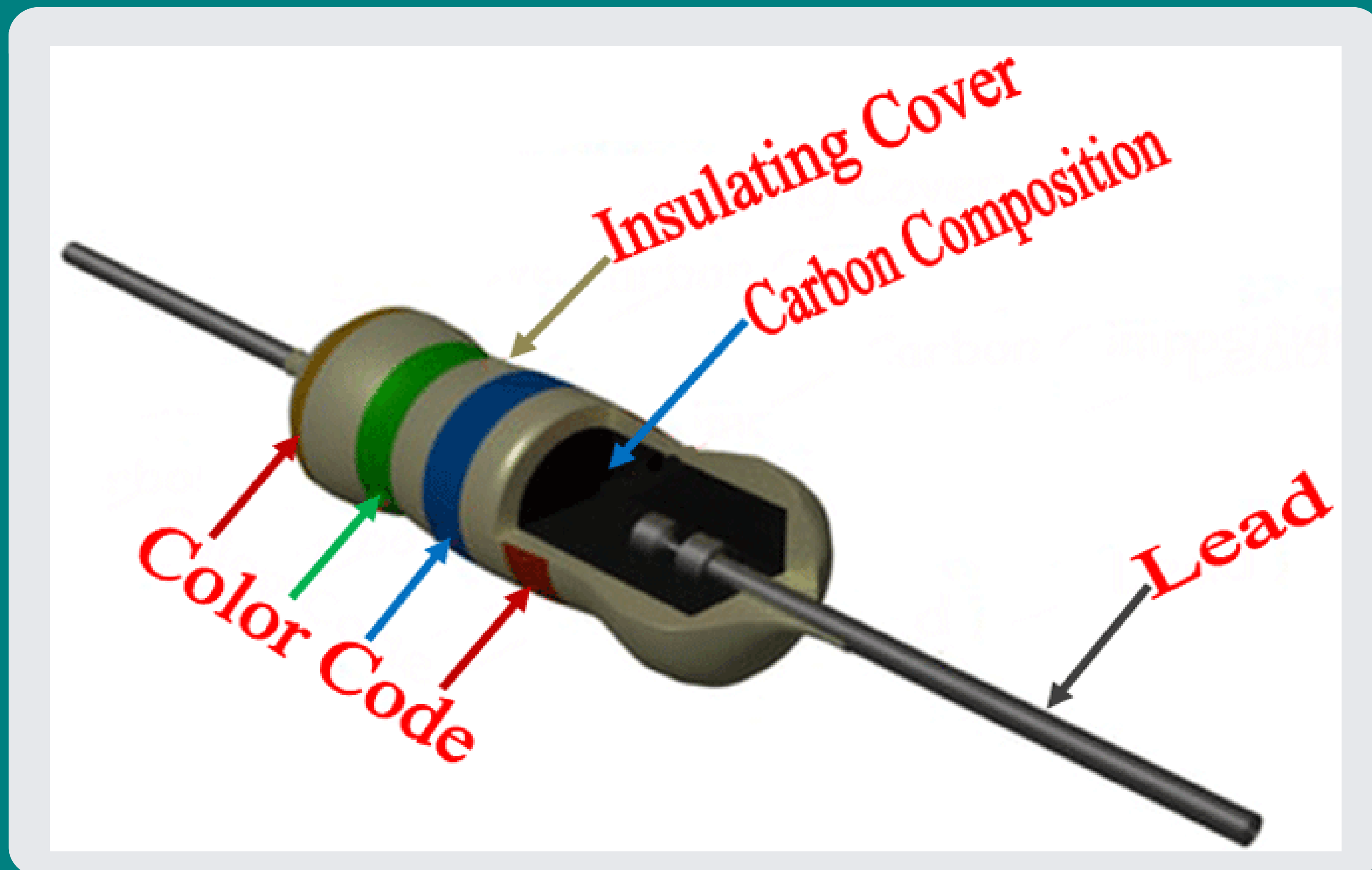


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① **Wire bound resistors :**

- They are made by winding the wires of an alloy, viz., manganin, constantan, nichrome or similar ones.
- They are **expensive, large**, but their **resistivity is insensitive to temperature**.

Types of Resistors



There are mainly **two types of Resistors** used in circuits and laboratories :

① **Wire bound resistors :**

② **Carbon resistors :**

- Made up of **Carbon compounds**.
- Carbon resistors are **compact, inexpensive** and thus used in small electronic circuits.
- Because of their **small size**, their **values** are given using a **colour code**.

Colour Codes for Carbon Resistors



- The first two rings give the first two significant figures of resistance in ohm.
- The third ring indicates the decimal multiplier.
- The last ring indicates the tolerance in per cent about the indicated value. Eg.
 $AB \times 10^C \pm D\% \Omega$
- Short Trick : **B B ROY of Great Britain has Very Good Wife**

Colour Codes for Carbon Resistors



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$$\rho_T = \rho_0 [1 + \alpha(T - T_0)] \quad (1)$$

where, $\rho_T \implies$ **Resistivity at a temperature T**

$\rho_0 \implies$ **Resistivity at reference temperature T_0 .**

$\alpha \implies$ **Temperature coefficient of resistivity.**

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- The above **linear relation** between resistivity (ρ) and Temp. (T) eq(1) does **not hold for all material**.
- Some materials like **Nichrome** (which is an alloy of nickel, iron and chromium), **Manganin and constantan** (used in wire bound standard resistors) exhibit a very **weak dependence of resistivity with temperature**

TEMPERATURE DEPENDENCE OF RESISTIVITY

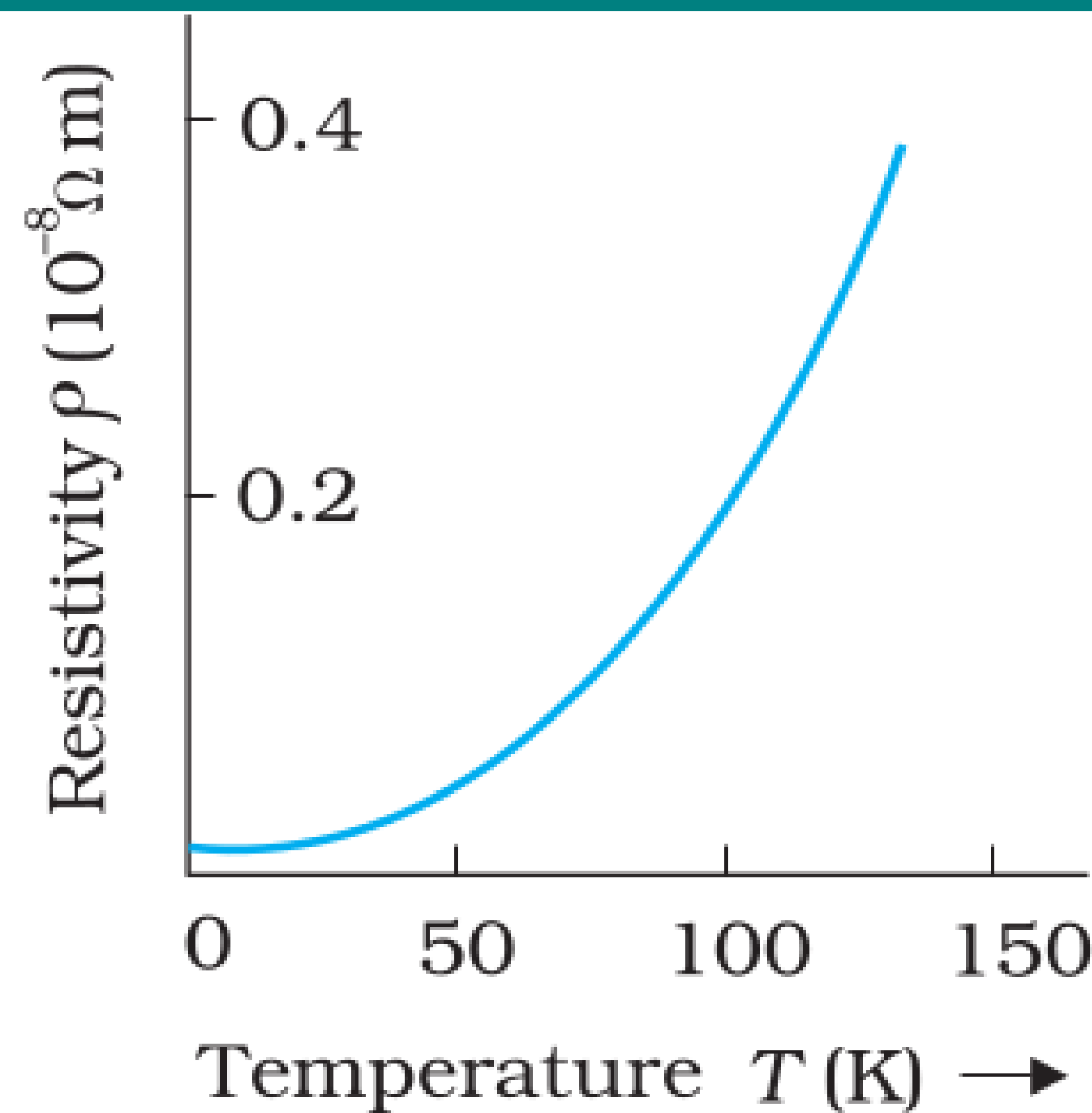


FIGURE 3.9
Resistivity ρ_T of copper as a function of temperature T .

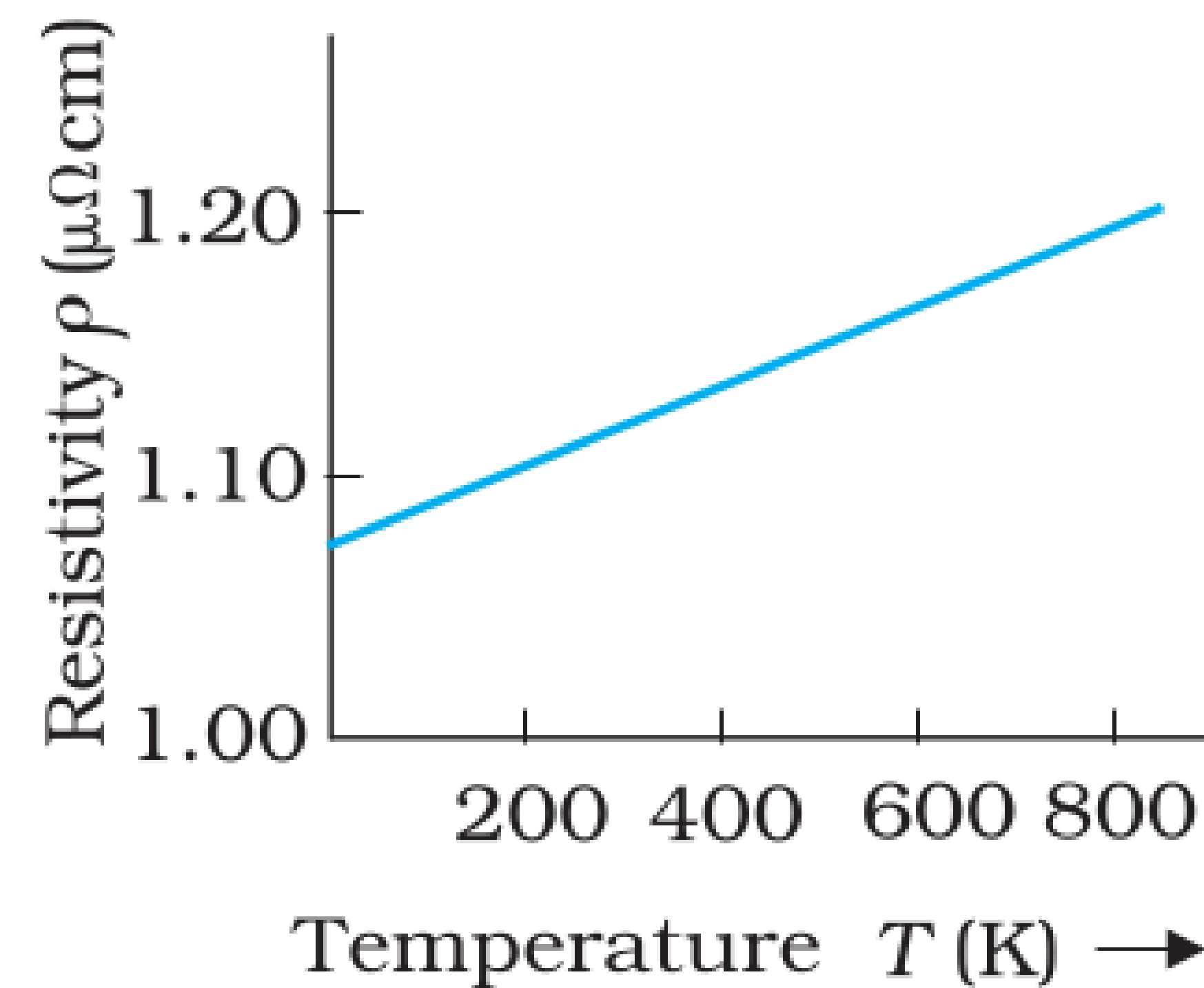


FIGURE 3.10 Resistivity ρ_T of nichrome as a function of absolute temperature T .

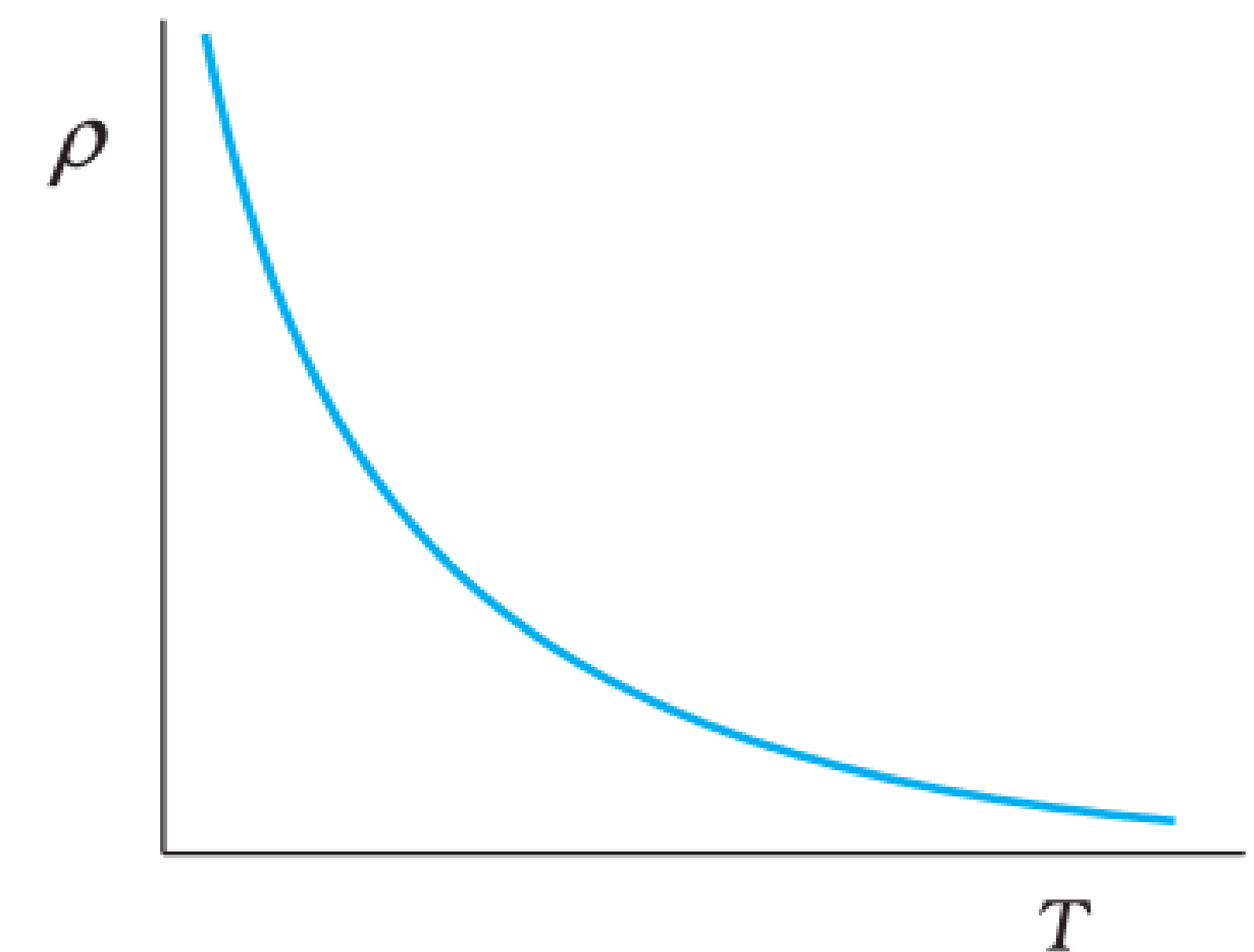


FIGURE 3.11
Temperature dependence of resistivity for a typical semiconductor.

Why resistivity ρ depends on temperature (T)

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- ρ thus **depends inversely** both on the **electron no. density n** and on the **relaxation time τ**
- As we **increase temperature**, **average speed of the electrons increases** resulting in **more frequent collisions**. Thus τ **decreases with temperature**.
- In a metal, **resistivity increases with temperature**.
Reason -relaxation time τ **decrease with rise in temperature** but n is **not dependent on temperature** to any appreciable extent.

Why resistivity ρ depends on temperature (T)

- **Resistivity (ρ)** of a material is given by-

$$\rho = \frac{m}{ne^2\tau} \quad (4)$$

- For **insulators and semiconductors resistivity (ρ) decreases with temperature.**
Reason - number of conduction electrons per unit volume n increases with temperature. This increase more than compensates any decrease in τ .

Example 3.4

The resistance of the platinum wire of a platinum resistance thermometer at the ice point is $5\ \Omega$ and at steam point is $5.23\ \Omega$. When the thermometer is inserted in a hot bath, the resistance of the platinum wire is $5.795\ \Omega$. Calculate the temperature of the bath.

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Solution:

Given :

Resistance (R_0) at ice Point ($T_0 = 0^\circ C$) = $5\ \Omega$

Resistance (R_{100}) at steam Point ($T_{100} = 100^\circ C$) = $5.23\ \Omega$

Resistance (R_T) at hot bath (T) = $5.795\ \Omega$

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Step 1 : First we calculate the temp. of coefficient of resistivity α

$$R_T = R_0 [1 + \alpha(T - T_0)]$$

$$R_{100} = R_0 [1 + \alpha(T_{100} - T_0)]$$

$$5.23 \Omega = 5 \Omega [1 + \alpha(100^\circ C - 0^\circ C)]$$

$$5.23 = 5 + 500\alpha^\circ C$$

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Solution:

$$\alpha = \frac{5.23 - 5}{500 \text{ } ^\circ\text{C}} = \frac{0.23}{500 \text{ } ^\circ\text{C}} = 0.00046 \text{ } ^\circ\text{C}^{-1}$$

$$\alpha = 4.6 \times 10^{-4} \text{ } ^\circ\text{C}^{-1} \quad (5)$$

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Step 2: Now, using this value of α calculating the temp. of the bath.

$$\begin{aligned} R_T &= R_0 [1 + \alpha(T - T_0)] \\ 5.795 \text{ } \Omega &= 5 \text{ } \Omega [1 + 4.6 \times 10^{-4} \text{ }^\circ\text{C}^{-1}(T - 0^\circ\text{C})] \\ \frac{5.795}{5} &= 1 + 4.6 \times 10^{-4} \text{ }^\circ\text{C}^{-1}(T) \end{aligned}$$

Example 3.4

Solution:

$$\begin{aligned}1.159 &= 1 + 4.6 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}(T) \\1.159 - 1 &= 4.6 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}(T) \\ \frac{0.159}{4.6 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}} &= (T) \\ T &= 0.03456 \times 10^4 \text{ }^{\circ}\text{C} \\ T &= 345.6 \text{ }^{\circ}\text{C}\end{aligned}$$

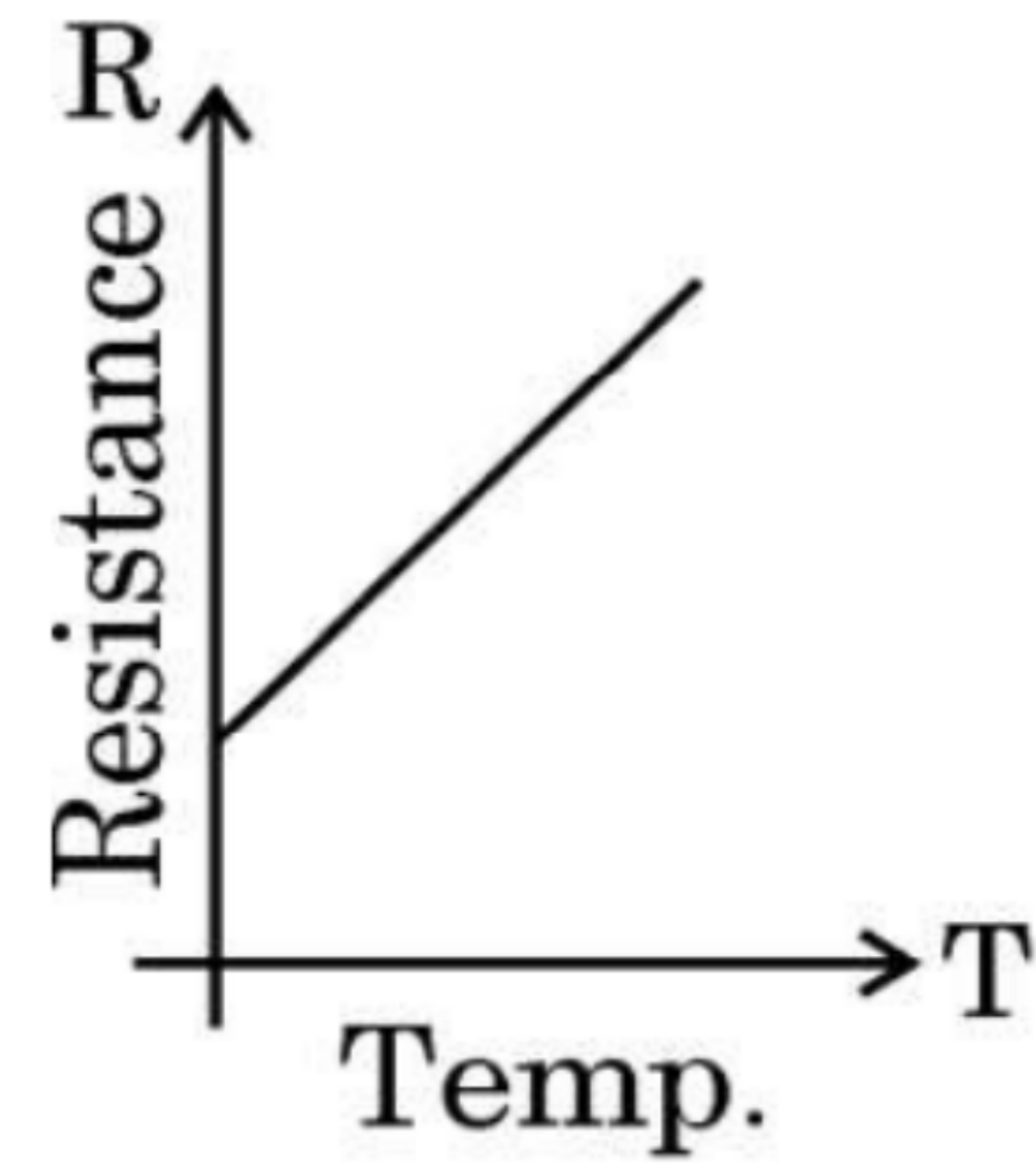
(6)

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T . The conductivity of

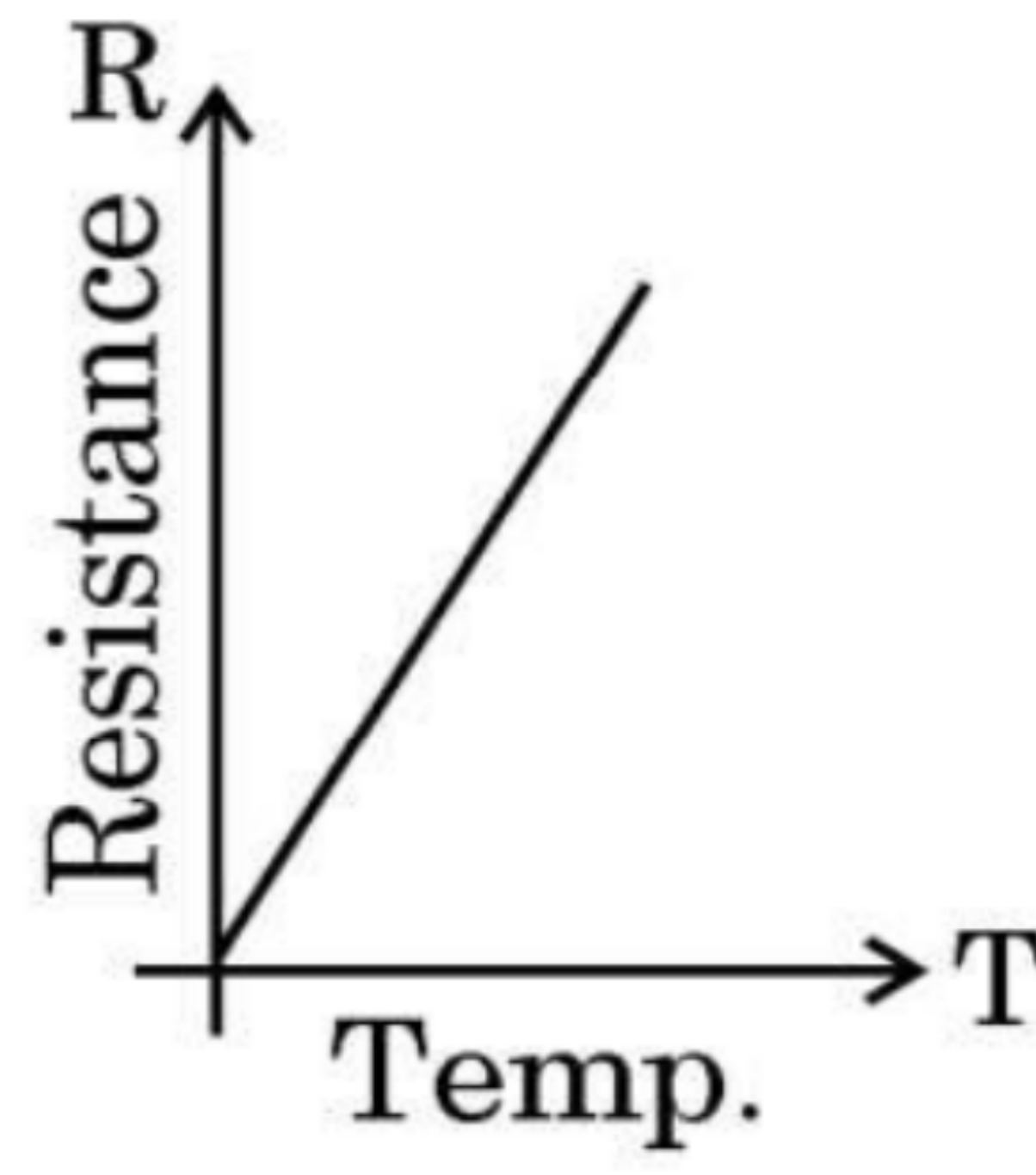
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- (a) both increases.
- (b) both decreases.
- (c) copper increases and silicon decreases.
- (d) copper decreases and silicon increases.

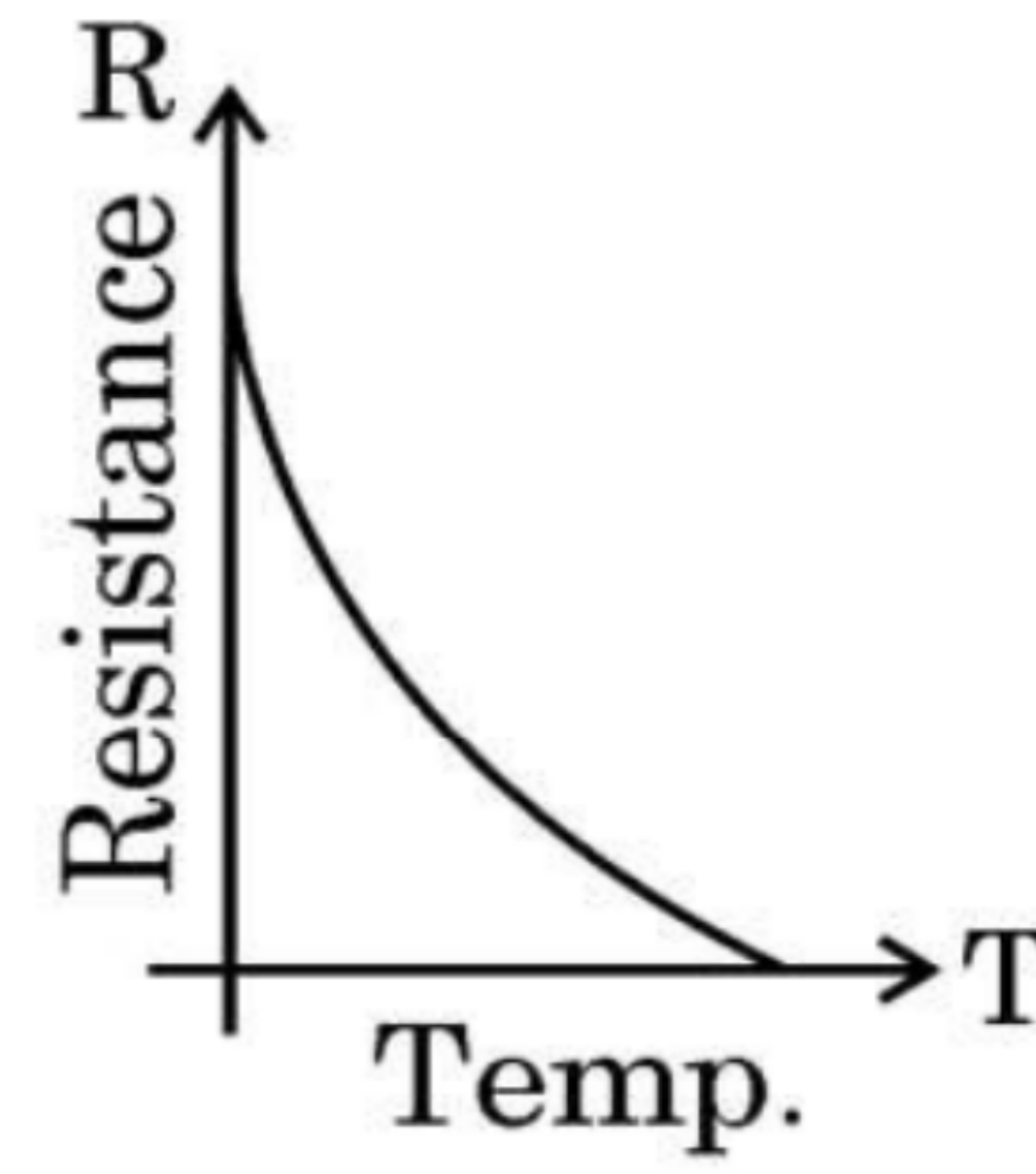
2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is :



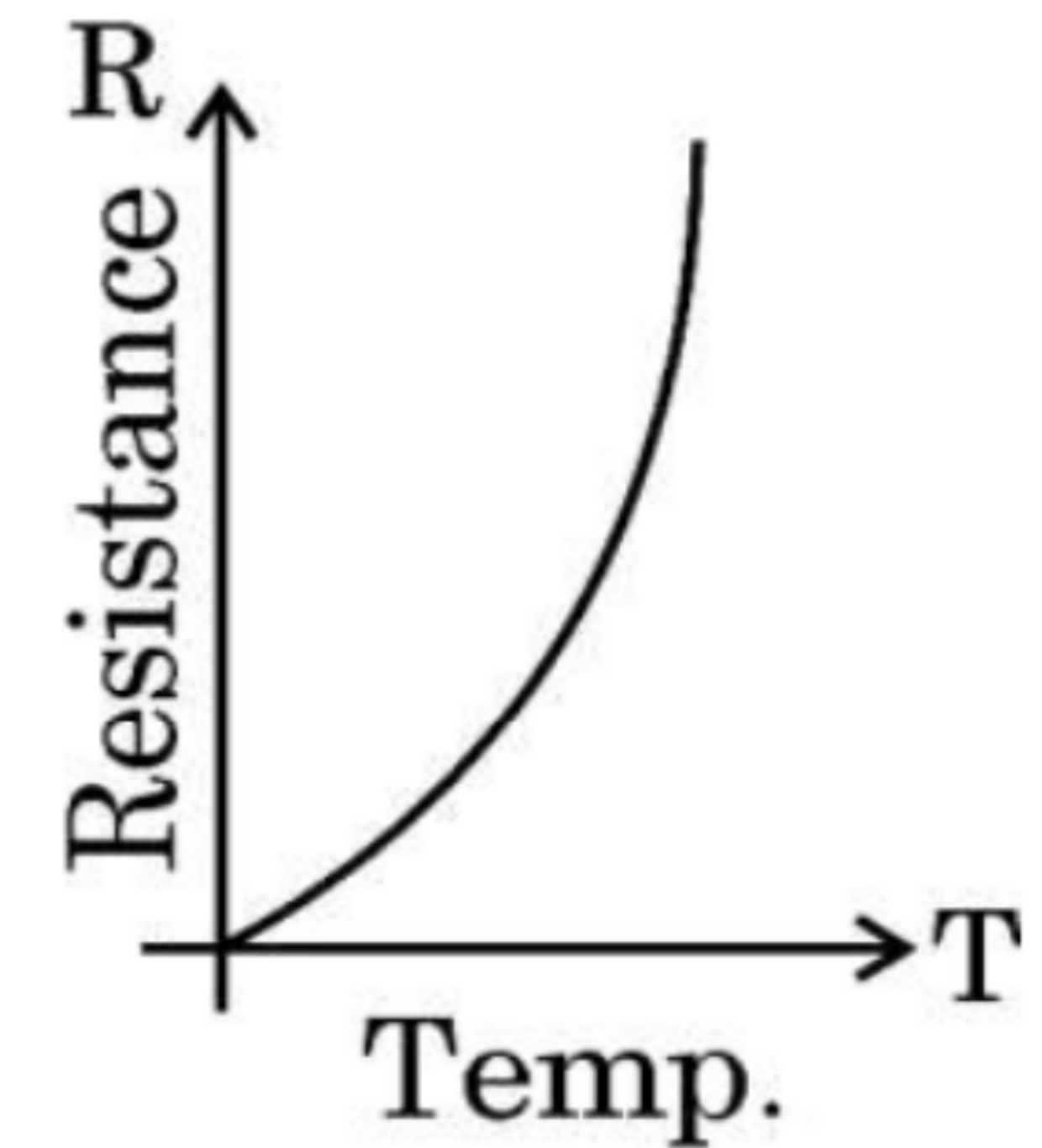
(a)



(b)



(c)



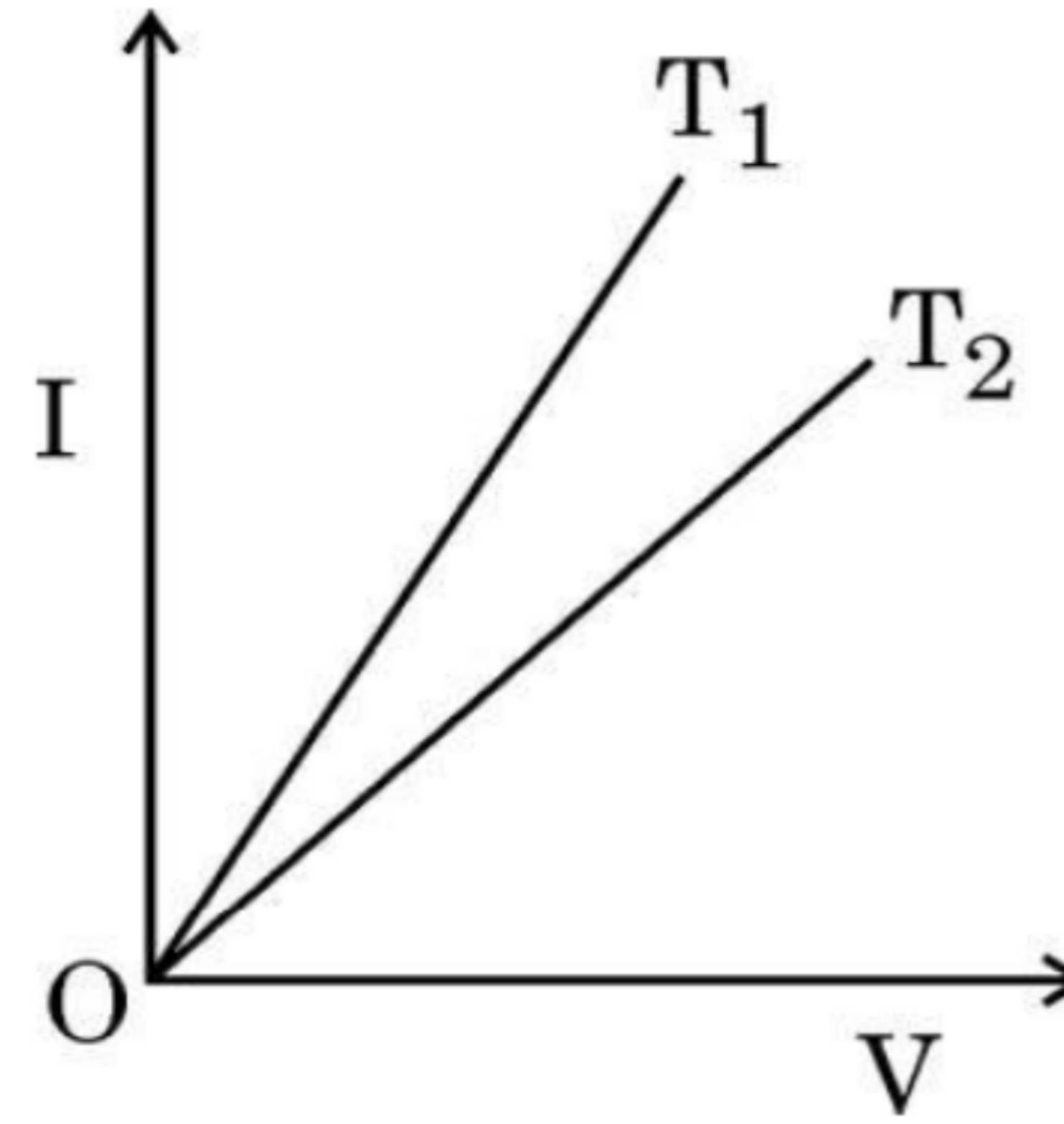
(d)

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

24. (i) Define 'temperature coefficient of resistance' of a metal.
- (ii) Show the variation of resistivity of copper with rise in temperature.
- (iii) The resistance of a wire is $10\ \Omega$ at $27\ ^\circ\text{C}$. Find its resistance at $-73\ ^\circ\text{C}$.
The temperature coefficient of resistance of the material of the wire is $1.70 \times 10^{-4}\ ^\circ\text{C}^{-1}$.

1. The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 and T_2 . One can conclude that :



(A) $T_2 = 2T_1$

(B) $T_1 > T_2$

(C) $T_1 = T_2 / 3$

(D) $T_1 < T_2$

2. The resistance of a wire at $20\text{ }^{\circ}\text{C}$ is $50\ \Omega$. When it is heated to $120\text{ }^{\circ}\text{C}$, the resistance becomes $51\ \Omega$. The temperature coefficient of resistance of the material of the wire is **1**
- (A) $2 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$ (B) $1 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$
(C) $5 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$ (D) $1 \times 10^{-3}\text{ }^{\circ}\text{C}^{-1}$

20. Find the temperature at which the resistance of a conductor increases by 25% of its value at 27 °C. The temperature coefficient of resistance of the conductor is $2.0 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$.

17. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°C. Take 20°C as the reference temperature and temperature coefficient of resistance of silver at 20°C = $4.0 \times 10^{-3} \text{ K}^{-1}$.

(ii) The temperature coefficient of resistance of nichrome is $1.70 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$. In order to increase resistance of a nichrome wire by 8.5%, the temperature of the wire should be increased by : 1

(A) 250°C

(B) 500°C

(C) 850°C

(D) 1000°C

(i) (a) Consider the contribution of the following two factors I and II in resistivity of a metal :

I. Relaxation time of electrons

II. Number of electrons per unit volume

The resistivity of a metal increases with increase in its temperature because :

(A) I decreases and II increases.

(B) I increases and II is almost constant.

(C) Both I and II increase.

(D) I decreases and II is almost constant.

OR

Example 3.3

An electric toaster uses nichrome for its heating element. When a negligibly small current passes through it, its resistance at room temperature ($27.0\text{ }^{\circ}\text{C}$) is found to be $75.3\ \Omega$. When the toaster is connected to a 230 V supply, the current settles, after a few seconds, to a steady value of 2.68 A . What is the steady temperature of the nichrome element? The temperature coefficient of resistance of nichrome averaged over the temperature range involved, is $1.70 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$.